Data Warehousing Project Configuration Management Plan

Data Warehousing Team
Data Warehousing Project
Alex Anthony
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Hayden Nikkel
[SCM Lead Signature Block]

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[Project Lead Signature Block]

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[Project Manager Signature Block]

Revision Sheet

Revision	Date	Brief Summary of Changes
Version 1.0(draft)	January 16th, 2026	Baseline document draft
Version 1.1(draft)	February 11th, 2026	Updated Introduction
Version 1.2(draft)	March 4th, 2026	Updated SCM configuration

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1. INTRODUCTION

[*This Configuration Management Plan template is designed to facilitate the definition of processes and procedures relating to software configuration management activities. This template was developed using IEEE Std 828-1998, IEEE Standard for Software Configuration Management Plans which has been adapted to support CMMI requirements.*

Information displayed in blocked text under each section is suggested items for inclusion in the Configuration Management Plan.

Information displayed in brackets is explanatory. Delete the bracketed text items and add your project-specific input. These items are food for thought on the section they address.]

1.1 Purpose

This plan exists to help plan configuration management for each person within the Data Warehousing team. It will be used to help with planning, identification (defining components/baselines), change control (tracking modifications), status accounting (recording history), and auditing.

1.2 Scope

This document defines Configuration Management (CM) activities for all Software and Data produced during the development of the Data Warehousing Project (DW Project) software. This document applies to all module products, end-user products, and data developed and maintained for the DW Project Program. CM activities as defined herein will be applied to all future DW Project projects.

This document conforms to Data Warehousing Team's Software Configuration Management Policy #1 and IEEE standards for software configuration management, and will change as needed to maintain conformance.

1.3 Definitions and Acronyms

1.3.1 Key acronyms

BCR -	Baseline Change Request
CCB -	Configuration Control Board
CER -	Change/Enhancement Request
CM -	Configuration Management
CRC -	Change Request Coordinator
DT&E -	Developmental Test & Evaluation

FOT&E -	Final Operational Test and Evaluation
ITR -	Internal Test Report
PL -	Project Lead
PM -	Program Manager
PCVS -	PC based source code version control software
SCM -	Software Configuration Manager
SCR -	Software Change Request
SE -	Software Engineer
SEPG -	Software Engineering Process Group
SPR -	Software Problem Report
SQA -	Software Quality Assurance
SRS -	Software Requirements Specification
TPR -	Test Problem Report (DT&E)
VDD -	Version Description Document

1.3.2 Key terms

The following key terms are used as defined in the IEEE Guide for the Use of IEEE Standard Dictionary.

configuration management	"A discipline applying technical and administrative direction and surveillance to: identify and document the functional and physical characteristics of a configuration item, control changes to those characteristics of a configuration item, control changes to those characteristics, record and report change processing implementation status, and verify compliance with specified requirements." [IEEE90].
customer	"The individual or organization that specifies and accepts the project deliverables. The customer may be internal or external to the parent organization of the project, and may or may not be the end user of the software product. A financial transaction between the customer and developer is not necessarily implied." [IEEE87]
database	"A collection of data fundamental to a system." [IEEE91]
installation	"The period of time in the software life cycle during which a software product is integrated into its operational environment and tested in this environment to ensure that it performs as required." [IEEE91]
plan	"A detailed scheme, program, or method worked out beforehand for the accomplishment of an objective." [Heritage85]
process	"A sequence of steps performed for a given purpose." [IEEE90]

project	-- unit of work to meet a specific customer requirement. Includes all tasks, activities, and functions necessary to meet the requirements.
project deliverables	"The work product(s) to be delivered to the customer. The quantities, delivery dates, and delivery locations are specified in the project agreement." [IEEE87]
quality assurance	"(1) A planned and systematic pattern of all actions necessary to provide adequate confidence that an item or product conforms to established technical requirements." [IEEE90] "(2) A set of activities designed to evaluate the process by which products are developed or manufactured." [IEEE90]
review	--A process or meeting during which a work product, or set of work products, is presented to program personnel, managers, users, customers, or other interested parties for comment or approval. Types include requirements review, design review, code review, test readiness review, formal qualification review.
software	"Computer programs, procedures, and associated documentation and data pertaining to the operation of a computer system." [IEEE90]
software approach **engineering**	"The application of a systematic, disciplined, quantifiable to the development, operation, and maintenance of software; that is, the application of engineering to software." [IEEE90]
software life cycle	"A project-specific, sequenced mapping of activities." [IEEE91]
software quality assurance	--See quality assurance.
specification	"A document that specifies, in a complete, precise, verifiable manner, the requirements, design, behavior, or other characteristics of a system or component, and, often, the procedures for determining whether these provisions have been satisfied." [IEEE90]
walk-throughs	"A static analysis technique in which a designer or programmer leads members of the development team and other interested

parties through a segment of documentation or code and the participants ask questions and make comments about possible errors, violations of development standards, and other problems."[IEEE90]

work product --Any tangible item that results from a project function, activity, or task. Examples of work products include customer requirements, project plan, design documents, source and object code, user's manuals.

1.4 References

IEEE87 IEEE Std 1058.1-1987, IEEE Standard for Software Project Management Plans

IEEE88 IEEE Std 982.2-1988, IEEE Guide for the Use of IEEE Standard Dictionary.

IEEE90 IEEE Std 610.12-1990, IEEE Standard Glossary of Software Engineering Terminology (ANSI).

IEEE91 IEEE Std 1074-1991, IEEE Standard for Developing Software Life Cycle Processes.

Schach96 Classical and Object Oriented Software Engineering, Schach, 1996

Davis90 Davis, A, "Software requirements: Analysis & Specification," Prentice Hall, 1990.

Heritage85 The American Heritage Dictionary, Houghton Mifflin Publishers, 1985.

Paulk93 SEI Capability Maturity Model, Version 1.1, CMU/SEI-93-TR-24.

MIL498 MIL-STD-498, Military Standard Software Development and Documentation, 5 December 1994.

2. Software Configuration Management (SCM)

[SCM management information describes the allocation of responsibilities and authorities for SCM activities to organizations and individuals within the project structure. SCM management information should include three topics: the project organization(s) within which SCM is to apply, the SCM responsibilities of these organizations, and references to the SCM policies and directives that apply to this project.]

2.1 SCM Organization

Figure 3.1 provides an overview of Data Warehouse Project SCM organization.

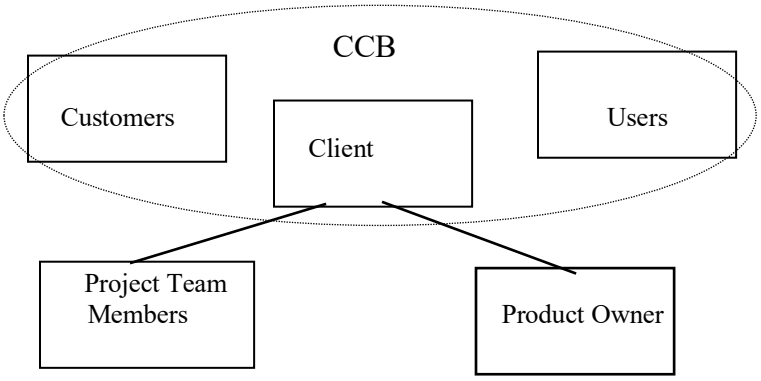

Figure 3.1. [Project Abbreviation] SCM Organization

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2.2 SCM Responsibilities

Scrum Master. The Scrum Master is responsible for ensuring the SCM process is developed, maintained and implemented. The Scrum Master is responsible for ensuring that XP programmers and the Product Owner are adequately trained in SCM policy and procedure. The Data Warehouse Project Scrum Master and Product Owner work together to develop, maintain, and implement effective software configuration management. Please refer to Figure 3.1 and 3.1.

The Data Warehouse Team Scrum Master has the authority to ensure development of an appropriate configuration management process. Since CCB responsibility exists with the customer (See Figure 3.1), team member(s) of the Data Warehouse Team Project will be identified as members of this CCB and will represent our interests there.

The Scrum Master is also responsible for implementing the actions of this process and for the implementation of the client's process improvement recommendations.

The Scrum Master ensures that the process is implemented once approved and that proposed changes to the process are reviewed by the Data Warehouse Project team members and approved as required.

The Scrum Master is responsible for adhering to the Data Warehouse Project CM process and for ensuring that only approved changes are entered into the module product baselines. The Scrum Master is also responsible for identifying the module version to be included in the end-user products. The Scrum Master is responsible for controlling the software product(s) development and/or maintenance baseline to ensure that only approved changes are entered into the baseline(s) system. The Scrum Master maintains the software product with all changes to be approved by the associated CCB.

The Scrum Master develops and proposes a baseline version of the software configuration management product, ensuring that the SCM process is followed, once approved, and proposes process changes to the Data Warehouse Project SCM and CCB, as required.

The Scrum Master is responsible for ensuring that project configuration management processes are followed. The Scrum Master also ensures all members of the project team are trained in SCM policies and procedures that apply to their function on the team. This consists of initial training for new members and refresher training, as required, due to process or personnel changes.

2.3 Relationship of CM to the software process life cycle

[This section should describe CM as it relates to other elements of the software process life cycle.]

2.4 SCM Resources

[This section should describe the minimum recommended resources (eg., time, tools, materials) dedicated to software configuration management activities.]

3. SCM ACTIVITIES

3.1 Configuration Identification

[All technical and managerial SCM activities should be described, as well as all functions and tasks required to manage the configuration of the software. General project activities that have SCM implications should also be described from the SCM perspective. SCM activities are traditionally grouped into four functions: configuration identification, configuration control, status accounting, and configuration audits and reviews. An example of the type of information provided in this section follows:

All [Project Abbreviation] Modules and baselines are to be approved by the CCB. Once a module has been baselined all changes made will be identified in their respective Version Description Documents (VDDs).

The [Project Abbreviation] is currently in development, product baseline to be approved upon delivery.

Product baselines are to be maintained by the [Project Abbreviation] SCM. The Baseline will consist of the following:

- source files
- libraries
- executable files
- data files

- makefiles
- link files
- VDDs
- User's Manuals

]

3.1.1 Specification Identification

[SCM activities should identify, name, and describe all code, specifications, design, and data elements to be controlled for the project. Controlled items may include executable code, source code, user documentation, program listings, databases, test cases, test plans, specifications, management plans, and elements of the support environment (such as compilers, operating systems, programming tools, and test beds).]

3.1.2 Change Control Form Identification

The Plan should specify the procedures for requesting a change. As a minimum, the information recorded for a proposed change should contain the following:

a) The name(s) and version(s) of the CIs where the problem appears;
b) Originator's name and organization;
c) Date of request;
d) Indication of urgency;
e) The need for the change;
f) Description of the requested change.

Additional information, such as priority or classification, may be included to assist in its analysis and evaluation. Other information, such as change request number, status, and disposition, may be recorded for change tracking. An example of the type of information provided in this section follows:

Software Change Request (SCR) procedures for each development team are defined in Appendix A. These procedures are used to add to, change, or remove items from the baselines. The identification and tracking of change requests is accomplished through Change/Enhancement Requests (CER).]

3.1.3 Project Baselines

[This section should describe the configuration control activities required implement changes to baselined CIs. The Plan should define the following sequence of specifc steps:

a) Identification and documentation of the need for a change;
b) Analysis and evaluation of a change request;
c) Approval or disapproval of a request;
d) Verification, implementation, and release of a change.

The Plan should identify the records to be used for tracking and documenting this sequence of steps for each change.]

3.1.4 Library

[This section should identify the software configuration management repository, if one exists. The following provides an example of the type of typical information:

[Project Abbreviation] is composed of end-user products, module products and data. The organization of these computer software units is given in Appendix C.]

3.2 Configuration Control

3.2.1 Procedures for changing baseline

[This section should describe the procedures for making controlled changes to the development baseline. The following provides an example of this type of information:

Changes to each software product baseline are made according to the change process.

Change Tracking. Changes to each software product baseline are tracked using an automated CER and SCR tracking system. These systems report current status for all CERs and SCRs for a specified version and product.

Change Release. The PL provides reports containing the following information for review at each CCB meeting for:

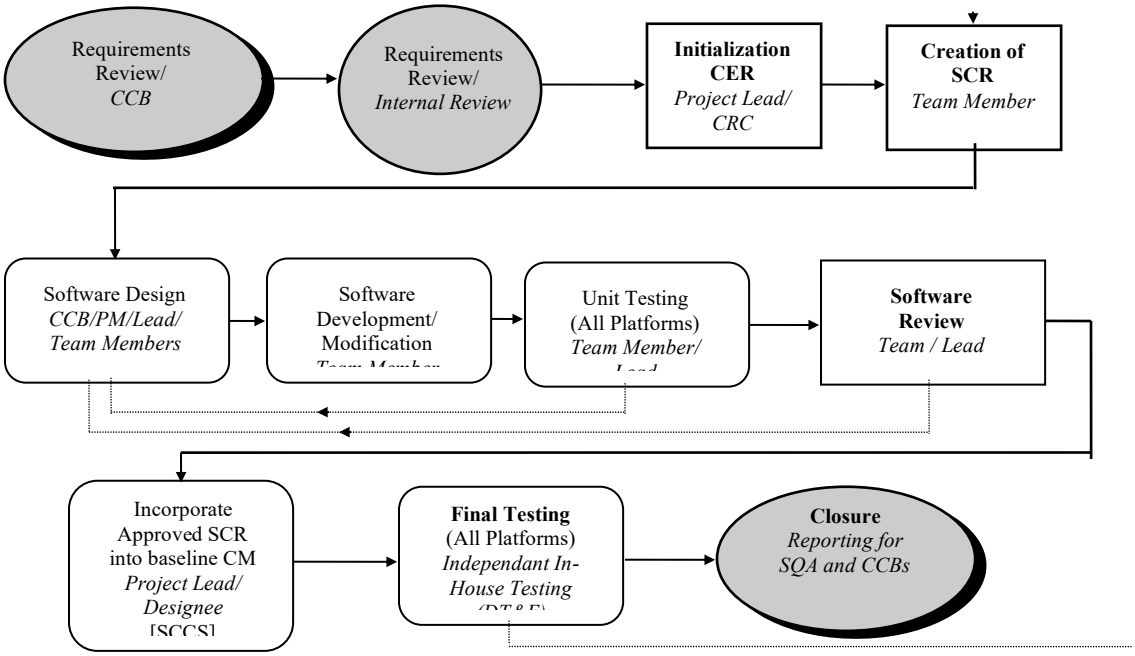
a) Number of new CERs
b) Type
c) Priority
d) Source (Internal, Field, FOT&E)
e) Action Required
f) Number of CERs in each status

Each development team Project Lead, or designated Change Request Coordinator (CRC), is responsible for keeping the information in the CER tracking system up to date so the information is available to the PL and PM prior to CCB meetings.]

3.2.2 Procedures for processing change requests and approvals

[This section should describe the procedures for processing changes requests and the associated approval process. The foolowing provides an example of this process:

All requests for changes to baseline items within each project follows the Software Change Request procedures shown in Appendix A. These procedures are outlined below for [Project Abbreviation] maintenance and production lifecycles:



SQA actitivities are applied Baseline Change Process. Please refer to [Project Abbreviation] SQA documentation.
PVCS - Source Code Control Sysem.

Figure 4-1 Software Baseline Change Process

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3.2.3 Organizations assigned responsibilities for change control

[This section should describe all organizations associated with the change control process and their associated responsibilities.]

3.2.4 Change Control Boards (CCBs)

[This section should describe the role of the CCB and refer to any associated documentation (eg., CCB Charter).]

3.2.5 Interfaces

[This section should describe any required software interfaces and point to the source of their SCM documentation.]

3.2.6 Level of control

[This section should describe the levels of control and implementing authorities. The following is an example:

Evaluating Changes. Changes to existing software baselines are evaluated by the software product's associated CCB.

Approving or Disapproving Changes. Changes to existing software baselines are either approved or disapproved by the associated project CCB.

Implementing Changes. Approved changes are incorporated into the product baseline by the development team as identified in Appendix A.]

3.2.7 Document revisions

[This section should describe SCM plan maintenance information . All activities and responsibilities necessary to ensure continued SCM planning during the life cycle of the project should be described. The Plan should describe:

a) Who is responsible for monitoring the Plan;
b) How frequently updates are to be performed;
c) How changes to the Plan are to be evaluated and approved;
d) How changes to the Plan are to be made and communicated.]

3.2.8 Automated tools used to perform change control

[This section should identify all software tools, techniques, equipment, personnel, and training necessary for the implementation of SCM activities. SCM can be performed by a combination of software tools and manual procedures. For each software tool, whether developed within the project or brought in from outside the project, the Plan should describe or reference its functions and shall identify all configuration controls to be placed on the tool.]

3.3 Configuration Status Accounting

[Configuration status accounting activities record and report the status of a project's CIs. The SCM plan should describe what will be tracked and reported, describe the types of reporting and its frequency, and how the information will be processed and controlled.

If an automated system is used for any status accounting activity, its function should be described or referenced. The following minimum data elements shall be tracked and reported for each CI: its initial approved version, the status of requested changes, and the implementation status of approved changes. The level of detail and specific data required may vary according to the information needs of the project and the customer.]

3.3.1 Storage, handling and release of project media

[This section should describe the storage, handling and release of SCM products. The following is provided as an example:

The initial approved version of any [Project Abbreviation] software product is separately maintained by the PL until the approval of subsequent version of the product. Upon approval of the subsequent version the initial version is held in the project SCM library. This library is under configuration control.]

3.3.2 Information and Control

3.3.3 Reporting

[This section should describe the format, frequency, and process for reporting SCM product baseline status. Providing Configuration Items to testing may require a Test Item Transmittal Report. This Test Item report should identify the person responsible for each item, the physical location of the item, and item status. The following is an example of status report of requested changes:

The status report of requested changes is available to the PM at any time. A schedule status report may be provided to the CCB weekly and reported to [Company Name] senior management on a monthly basis. Please refer to the [Project Abbreviation] Software Development Plan, Section 6, for a status report schedule example.]

3.3.4 Release process

[This section should describe the release process, including all required approvals. A summary chart is sometimes used to describe the release process effectively.]

3.3.5 Document Status Accounting

[This section should describe how changes to the SCM plan are managed and reported.]

3.3.6 Change Management Status Accounting

[This section should describe the format, frequency, and process for reporting SCM change management status.]

3.4 Configuration Audits and Reviews

[This section should provide information in support of required configuration audits and reviews. The following is provided as an example:

Each team PL reviews the software baseline semi-annually. The [Project Abbreviation] PM and [Project Abbreviation] SCM review the baseline of each software product annually. This baseline review consists of checking for CER implementation. The CERs are randomly selected. Discrepancies will be annotated and reported to the PL. Corrective action is taken if required.

The [Project Abbreviation] CCB reviews the SCM process annually. Randomly selected problem reports are identified and walked-through the change process. Discrepancies are annotated and reported to the [Project Abbreviation] PM and [Project Abbreviation] SCM. Corrective action is taken if required.

A software baseline, the associated status reporting, and associated documentation for each software product is available for SQA review at all times.]

4. CM Milestones

[This section should provide a description of the minimum set of SCM related project milestones that are acceptable for compliance. These milestones should be reflected in the Software Project Management Plan schedule and resource allocation.]

5. Training

[This section can refer to an independent training plan. However, this training plan must include information relating specifically to SCM training. If an independent plan does not exist, then information regarding the type and frequency of SCM training should be identified here.]

6. Subcontractor/Vendor Support

[This section should describe the SCM activities required by subcontractors. The following information is provided as a supporting example:

[Project Abbreviation] subcontractor support will be identified by the [Project Abbreviation] Program Manager. Subcontractor support will be required to follow the process defined in this document. If a wavier is requested, the subcontractor must provide evidence of comparable configuration management procedures. These procedures will follow the same audit and control procedures described in Section 4.4.]

APPENDIX A // Software Change Request Procedures

A change to [Project Abbreviation] software may be requested by one of the following: Internal Test Report (ITR), Software Problem Report (SPR), or Baseline Change Request (BCR) from external or internal sources or from the product specification/requirements. A change to [Project Abbreviation] documentation may be requested by a Document Change Request (DCR). All of the above requested change procedures are referred to as a change request in the text below.

The Configuration Control Board (CCB) for each development team determines which of the change requests is required for each software release prior to the start of work for that release. Additional change requests are reviewed by [Project Abbreviation] CCB to determine and assign the proper status to the change requests, these are held for CCB scheduling. Status is one of the following:

Open	-	Change request is to be implemented for current software version.
Hold	-	Change request targeted for another software version.
Voided	-	Change request is a duplicate of an existing CER or does not apply to existing software.
Working	-	Change is currently being implemented.
Testing	-	Implemented change request is under DT&E evaluation.
Fixed	-	Change request is implemented, unit and integration test complete.

The following steps define the procedure for each status. Each procedure starts with the change request being given to the Project Lead or designated Change Request Coordinator (CRC).

A. Open

1. INITIALIZATION

a) The CRC for the associated [Project Abbreviation] project creates a CER using the CER system.

b) The CRC notifies the Project Lead of the new CER.

2. CREATE SCR

a) The PL reviews the CER, and provides estimates of schedule impact to CCB.

b) CCB prioritizes CER and authorizes implementation.

c) The PL assigns the CER to a Software Engineer (SE) to incorporate.

d) The SE creates a Software Change Request (SCR).

3. SOFTWARE DESIGN

The SE designs changes necessary to implement CER:

a) The SE identifies interface changes and consults with appropriate PLs.

b) Determines changes needed.

c) The SE identifies any changes needed for documentation.

d) The SE estimates work effort required for CER completion.

4. SOFTWARE DEVELOPMENT/MODIFICATION

b) The SE makes needed changes and provides updated documentation for testing.

c) The SE checks out the files from the configuration managed version (not the baselined version) of all the files needed to incorporate the CER.

NOTE: [Project Abbreviation] utilizes [Software Configuration Tool Name] for configuration management. [Software Configuration Tool Name] keeps track of the product release versions of each source file and is used to checkin and checkout files. This product allows version independance, keeping each released version intact, allowing for release backdating.

d) The SE fills in the CM sections of the SCR.

e) The SE puts the changes in the configuration managed files.

5. UNIT TESTING

The SE tests on all platforms, if test fails go to Step 3, Software Design.

6. SOFTWARE REVIEW

a) The SE prepares a software review package:
    1) Hard-copy of SCR.
    2) List of documentation changes.

b) Software Review
    1) SE gives software review package to another team member for review.
    2) If SCR peer approved go to step 7.
    3) SCR not approved go to step 3, Software Design.

7. INCORPORATE APPROVED SCR INTO BASELINE CM

a) The PL changes the SCR status to indicate that it is approved.

b) The SE checks in the files.

c) The SE fills in the CM sections of the SCR.

f) The CRC reviews SCR for completeness.

g) The CRC changes the status on the CER to indicate testing.

h) The designated SCM gives a list of changed files to PLs for updating their working directories/executables.

i) The designated SCM installs changes to baseline on all platforms.

8. FINAL TESTING

a) The CRC identifies all CERs with testing status.

b) Final testing by DT&E validation group (If problems are found start over at step 2a)

9. CLOSURE

The CRC changes final status on CER to indicate complete.

B. Hold

The CRC creates a CER using the CER system on the SUN marking status HOLD and indicates the targeted version for incorporating the change. The CRC files the original change request for the next version. The change will be considered for inclusion in the next version.

C. Voided

1. Duplicates:

   a) The CRC marks the change requests as a duplicate, specifying which CER is a duplicated.

   b) The CRC files the change request for associated version.

2. Other:

   a) The CRC marks the reason for cancellation.

   b) The CRC files the change request for associated version.

D. Testing

The CRC forwards CER/SCR to PL for review. After review the PL forwards CER to DT&E test for validation. CER status is updated to indicate testing status.

E. Fixed

The CRC files the closed change request and updates database.

Software Change Request Procedures

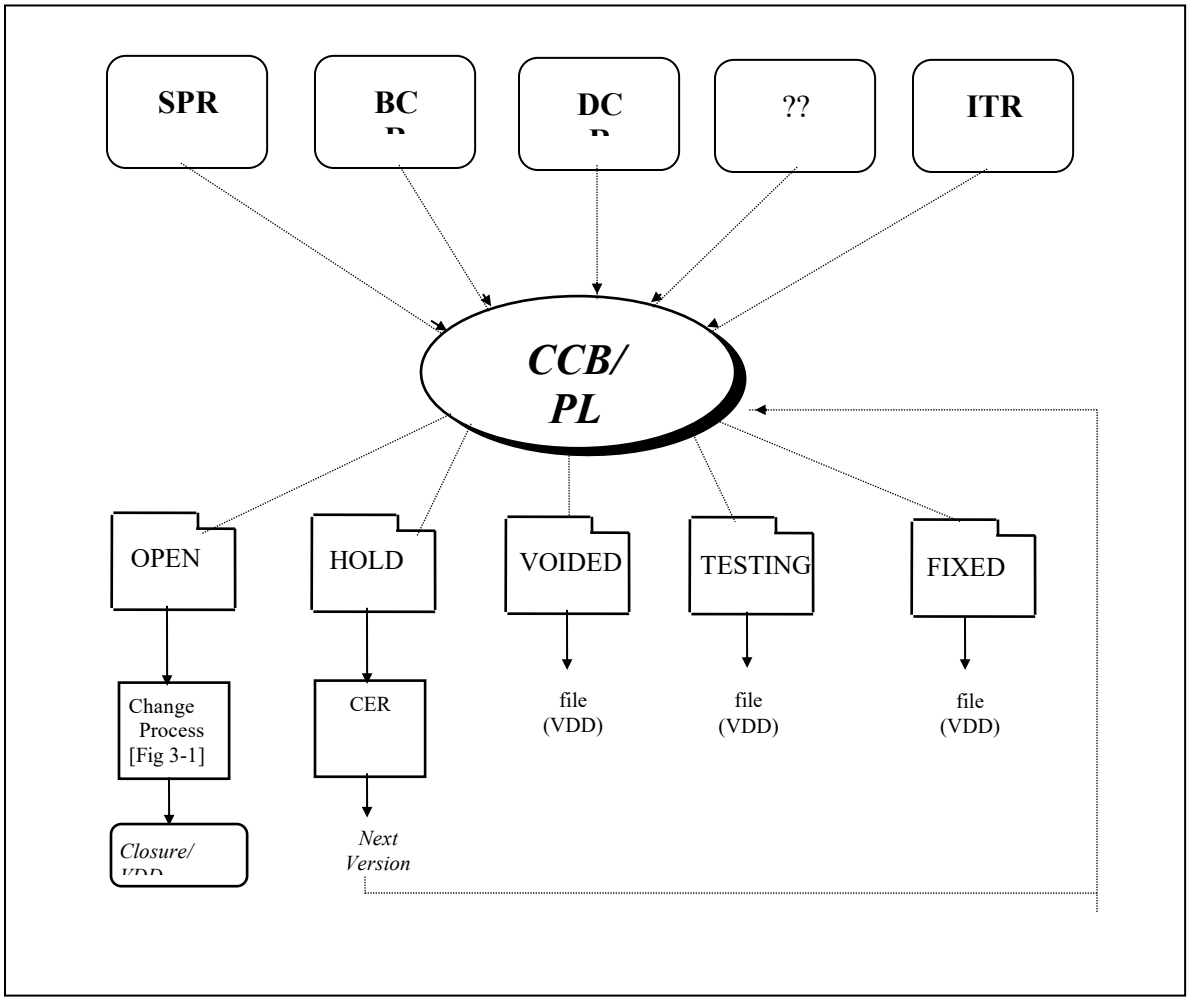

Figure A-1

APPENDIX B // [PROGRAM ABBREVIATION] Software Organization

G.1 [PROGRAM ABBREVIATION] Organization

[PROGRAM ABBREVIATION] software elements can be catagorized into three groups: module products, data products, and end-user products.

Module products and data products are used by software developers and not the users and are components of [PROGRAM ABBREVIATION].

Module products - Module products are software libraries that perform a specific set of functions. Module products provide re-usable code ensuring consistancy in function performance, eliminating duplication of effort in developing like functions, and reducing the amount of code. Module products are used in our end-user products and may be available for use by other companies/agencies.

Data products - Data products are database data files. Data products allow a single development, maintenance, and testing source and provides consistient data and format. Data products are included as part of our end-user products and may be available for use by other companies/agencies. Data products may be sent to users as a product itself.

End-user products - End-user products are executable programs for users.

Configuration management of module products is accomplished according to this document. Each module product has an individual baseline and version numbers. Module product managers specify the version of the module product to be used in end-user products.

G.1.1 [PROGRAM ABBREVIATION] Module Products

A PVCS directory structure for all module products exists for each software product version released.

G.1.2 [PROGRAM ABBREVIATION] Data Products

G.1.3 [PROGRAM ABBREVIATION] Deliverables

G.1.4 [PROGRAM ABBREVIATION] Support Software